

Kamal Kishor Sharma

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Profile

Problem-solver with eight years of combined industry and research experience across multiple disciplines spanning aerospace, mathematics, and computational physics. Skilled in translating advanced simulation results into validated engineering solutions, demonstrated through high-impact aerospace work at the Indian Space Research Organisation. Experienced in bridging mathematical modelling and practical implementation for real-world physical systems.

Core Competencies

- **Mathematical Modelling:** Proficient in modelling wide range of physical phenomena (structural, thermal, wave, fluid, and coupled systems) using mathematical equations
- **Numerical simulations:** Expertise in using numerical methods to solve mathematical models
- **Simulation/Modelling Software:** ANSYS (Workbench/APDL), Abaqus, Firedrake, AutoCAD, CATIA
- **Programming:** Python (NumPy, SciPy, PyTorch, Matplotlib), MATLAB; Git version control
- **Communication:** Skilled in scientific communication, presenting complex results to diverse audiences

Education

- PhD, Applied Mathematics** 2021 – Present
University of Hamburg & Max-Planck Institute for Meteorology, Hamburg Grade: 1.3
- Research Focus: Stochastic Partial Differential Equations (PDEs), Geophysical Fluid Dynamics, Model Development and Numerical Simulation
- MSc, Mathematical Modelling in Engineering (MathMods)** 2018 – 2020
University of Hamburg, Germany & University of L'Aquila, Italy Grade: 1.77
- Main modules: Scientific Computing, Advanced Modern Methods of Optimization and Approximation, Advanced Differential Equations
- BTech, Aerospace Engineering** 2010 – 2014
Indian Institute of Space Science and Technology, India

Experience

- Doctoral Researcher** Hamburg, Germany
University of Hamburg March 2021 – May 2025
- Developed a [stochastic climate model](#) (using an open-source Python Finite Element package) which can simulate fundamental interactions between the ocean and the atmosphere.
 - Built numerical solvers for geophysical fluid dynamics (GFD) models and conducted high resolution simulations.
 - Performed tests to quantify the uncertainty due to unresolved small-scale processes in GFD models.
 - Created new algorithms for calibration of stochastic models, improving predictive modelling.
 - Published research and presented results at five international conferences.
- Teaching Assistant** Hamburg, Germany
Department of Mathematics, University of Hamburg March 2024 – Sep 2024
- Taught a graduate-level course on [Mathematical Machine Learning](#).
 - Supervised projects on Physics-Informed Neural Networks (PINNs) and Large Language Models (LLMs).
- Structural Design & Analysis Engineer** India
Indian Space Research Organization (ISRO) Sep 2014 – Sep 2018
- Performed Finite Element Analysis (FEA) of critical [aerospace systems](#) (ranging from rocket components to satellite subsystems) for design validation and risk assessment.
 - Designed payload adapters, composite motor cases, and thermal screens, ensuring mechanical integrity under operational loads.
 - Supported hardware qualification campaigns, correlating thermal and structural simulations with test data.

Selected Engineering Projects

- **Scramjet Engine Flow Duct:** Conducted FEA of engine flow duct under aerodynamic, thermal, and inertial loads, ensuring structural integrity; supported hardware qualification testing.
- **Cryogenic Stage Thermal Screen:** Led design and analysis of a thermal screen for the cryogenic stage of India's heaviest and most powerful launch vehicle (GSLV Mk3).
- **Payload adapter for the Polar Satellite Launch Vehicle (PSLV):** Designed and validated a payload adapter for the PSLV; performed structural optimization for dynamic stability under launch loads.

Key Contributions

- **ISRO:** Conducted thermo-structural design, analysis, and testing for India's first scramjet engine and reusable launch vehicle.
- **PhD:** Developed a new discretization scheme and conducted the first ever numerical simulation of a coupled ocean-atmosphere stochastic climate model.

Machine Learning Exposure

- Completed several machine learning (ML) courses during my masters and PhD covering the mathematical and statistical concepts behind common ML techniques.
- Built a data-driven model based on convolutional neural networks to reconstruct computer tomography medical images.
- As a part of ML for climate science course, conducted literature review of ML models used for precipitation forecasting and presented a report on [MetNet-2](#) [↗](#) (a probabilistic weather model based on deep neural networks).

Language Proficiency

- English - Fluent | German - Intermediate (B1 level)

Scholarships

- Erasmus Mundus Scholarship (MSc) | Government of India Scholarship (BTech)